

IRSpectra-Bench and IRexp: candidate recall, not verification, limits LLM elucidation from real experimental IR and NMR

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Abstract. We release *IRexp*, the largest openly redistributable collection of experimental infrared band lists (121,233 records; 43,060 structure-linked; 33,201 full IR + ^1H + ^{13}C + structure quadruples), and *IRSpectra-Bench*, a blind mechanically scored benchmark of 194 compounds drawn from it. On *IRSpectra-Bench*, given the molecular formula and peak lists exactly as reported in open-access papers, a frontier large language model recovers the correct constitution for 28% (top-1; 95% CI 22–35), or 15% once reweighted to the corpus composition — far below the near-100% implied by curated demonstrations. The bottleneck is candidate proposal, not verification: the true structure enters the pool for only 34% of compounds, and where it does, training-free forward-verification selects it 89% of the time (58/65; 81% on the 37 compounds where a ranker had a choice). Recovered from published top- k figures, the same split carries 68–83% of the accuracy collapse when three systems move from curated or simulated data to real heterogeneous spectra. Generating wider lifts recall 32% $\rightarrow$ 42% and top-1 23% $\rightarrow$ 30% on 60 compounds; verification itself moves whole-benchmark top-1 only 28% $\rightarrow$ 30%. Masking the spectra drops top-1 from 23% to 5%, and Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol all verify better than they generate. All predictions and code are released.

1. Introduction

Determining structure from spectra is central to synthetic and analytical chemistry. Machine learning uses encoder-decoder models on paired corpora; *Spectro*, for one, learns $^1\text{H}/^{13}\text{C}/\text{IR} \rightarrow \text{SELFIES}$ from 6,833 molecules¹. General-purpose LLMs are now reported off-the-shelf: a non-peer-reviewed 2026 industrial white paper found Claude Opus matching or beating commercial NMR-prediction software forward (structure $\rightarrow$ spectrum, ± 0.08 ppm ^1H) and “recovering all eight simpler structures on every attempt” in the inverse². We treat those numbers as a claim to test, not an established baseline.

That evaluation is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, seven given the *starting-material structure* as a hint, and “recovery” scored leniently over three runs and three ranked candidates. The chemist’s question — *take an arbitrary experimental spectrum from a paper and recover the structure* — remains open. It needs open experimental data at scale, a blind benchmark others can report against, and honest accounting of which stage of elucidation binds.

We provide all three. *IRexp* (§2) is, to our knowledge, the largest openly *redistributable* collection of experimental IR *band lists* (121,233 records; 43,060 structure-linked; 33,201 IR + ^1H + ^{13}C + structure); view-only libraries such as SDBS hold more structure-linked *spectra* but cannot be redistributed (§2.1). *IRSpectra-Bench* (§3) is the blind, mechanically scored benchmark built from it — 194 compounds, complexity-stratified, multimodal IR + ^1H + ^{13}C — not the first such suite (MolQuest scores 530 post-2025 compounds by exact canonical SMILES³) but, to our knowledge, the first openly redistributable one on literature-reported peak lists with a recall/verification decomposition measured on Claude, replicated across three other model families (§4.7), with a within-compound solver-methodology control (§4). Ground truth is literature structures resolved deterministically (OPSIN/RDKit) and checked mechanically (560/560 bands on a seed-fixed sample, §2.3; expert-chemist review prepared but not yet run, §8); no LLM curates labels or scores predictions.

Forward-verification elucidation (§5) turns the model’s *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry); the loop is prior art (§1.1), the decomposition ours, run training-free on every compound. The finding is sharp asymmetry: given a candidate set containing it, the

model *verifies* the correct structure 89% of the time yet *proposes* it for only 34%, so generation binds the result (Fig. 1). Gain is bounded (top-1 28% $\rightarrow$ 30% on $n=194$; 23% $\rightarrow$ 30% on the 60-compound arm where wide generation was also tested) and, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

This paper contributes:

- **IRexp** — 121,233 experimental IR band lists, 33,201 full IR + ^1H + ^{13}C + structure quadruples, permissively licensed and redistributable.
- **IRSpectra-Bench** — a blind, mechanically scored 194-compound benchmark on literature-reported peak lists, complexity-stratified, with a within-compound control.
- **A recall/verification decomposition** on every compound: the true structure is proposed for 34% and, once proposed, selected 89% of the time.
- **A four-vendor replication** (Grok, Gemini, GPT, Claude) showing the same split, plus training-free and trained complements that move the ceiling rather than the protocol.

Solver and verifier (Fig. S1) are LLM agents on a consumer subscription, with no fine-tuning and no API spend. Inference is not exactly reproducible — no pinned snapshot, temperature or seed (§9) — but scoring is: predictions, ground truth and scorers are released, and every training-free number regenerates from them. The HOSE lookup, GNN (§5.4) and trained generator (§5.6) are fenced complements.

1.1 Related work

Trained spectra $\rightarrow$ structure models. The dominant line trains sequence or graph decoders on paired corpora — *Spectro*¹, multitask 1D-NMR models⁴, NMRTrans/NMRSpec⁵, *NMIRacle* (IR + ^1H + ^{13}C)⁶ — accurate in-distribution but retrained per modality. *IRexp* (§2.3) complements NMRSpec with redistributable IR band lists; neither corpus alone is multimodal. Upstream IR work from our group (*vIR-OLO*⁷, *j-IR-vis*⁸) identifies functional groups but does not generate candidate structures.

LLMs as elucidators. LLM agents already plan syntheses (ChemCrow⁹, Coscientist¹⁰) and call elucidation routines; we measure what one returns. Off-the-shelf and multimodal LLMs^{2,11–13}, puzzle benchmarks (MolPuzzle¹⁴), and agentic IR interpreters (IR-Agent¹⁵; Priessner *et al.*¹⁶) improve *how* a model reads spectra or re-ranks a fixed pool; we ask which stage binds once it has read them.

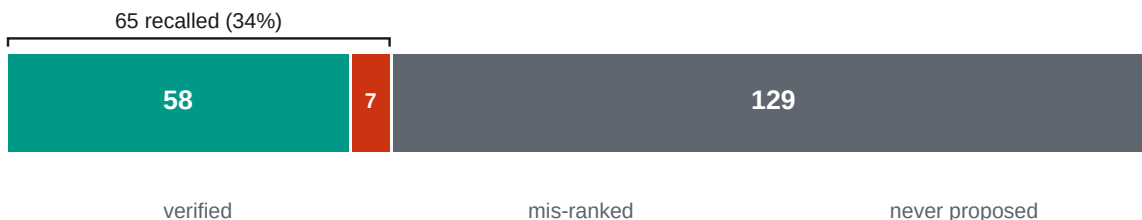


Fig. 1. Diagnosis on IRSpectra-Bench (n=194): generation recall, not verification, is the bottleneck. Where the true structure is never proposed, no ranker can recover it; where it is proposed, verification usually selects it. End-to-end top-1 is the product of those two rates, which have different denominators and are not differenced.

Most benchmarks use simulated or single-instrument spectra^{1,6,17,18}; IRSpectra-Bench uses literature-reported peak lists across thousands of laboratories. Recent suites bound the task elsewhere (NMRGym¹⁹, MolQuest³, SpecX²⁰). Cross-paper numbers swing with harness: GPT-4o scores 1.4% on MolPuzzle¹⁴, 27.8% with chain-of-thought, 57.8% with tree-search¹³. We fix one pre-registered RDKit-InChIKey protocol with bootstrap CIs.

Forward verification and CASE. Matching predicted to observed shifts — DP4^{21,22}, NMR crystallography^{23,24}, CASE^{25,26} — is easier than inverse generation. We swap DFT for a forward LLM. CASE enumerators have exhaustive recall by construction; the LLM literature almost never reports whether the true structure was proposed at all (§6).

Prior loops and concurrent work. *NMR-Solver*²⁷ runs the same generate-and-verify loop without an LLM (52.9% top-1 on literature ¹H/¹³C, formula supplied; 31.6% with stereochemistry). We decompose error rather than beat that score. *NMRAgent*²⁸ is a complementary best-effort system. Kliuev *et al.* place the best of six LLMs at 24% on 105 peak lists²⁹. Espejo Morales *et al.* frame agentic search with processing tools and hard checks: 80.9% on education spectra, 20.6% on 34 industrial samples³⁰ — they ask which architecture; we ask which stage binds. Their inputs are raw instrument files; ours are published reports (§8). Priessner *et al.* already state that re-ranking cannot recover a missing candidate¹⁶. We add measurement at scale: the split on 194 compounds across four model families and four verifiers, recovered from published top-k figures (§6).

2. The IRExp dataset

2.1 Motivation

An IRExp record is a *band list* — peak positions in cm⁻¹ transcribed by an author into a paper’s text — with co-reported ¹H/¹³C shift lists and, where resolvable, the 2D structure. IRExp contains no absorbance traces. That is the published form a language model consumes, but a different object from a digitised spectrum, and the two should not be counted against one another.

Among *digitised spectra*, free downloads run to the NIST WebBook³¹ (~16k) and the Chemotion ELN deposit³² (~2k). AIST SDBS³³ is larger (~54k FT-IR, all structure-linked) but *view-only* (50 spectra/day, no bulk export), so IRExp does not compete there: SDBS alone holds more structure-linked spectra than IRExp holds structure-linked band lists. Among text-derived *band lists* IRExp is the largest *openly redistributable* collection by record count — deliberately scoped to that object type.

2.2 Construction

Experimental sections report per-compound IR band lists with co-reported ¹H/¹³C NMR in a stable textual convention. Mining it is mature³⁴; we add a pipeline specialised to the *IR band list*, the modality those tools passed over.

- **Discovery.** Open-access literature is harvested in bulk from the PMC Open-Access Subset³⁵ and the Chemotion FT-IR deposit (RADAR4Chem).
- **Extraction.** A deterministic parser segments experimental text per compound and extracts IR wavenumbers and ¹H/¹³C shifts, gated against scan-range artefacts and prose false-positives.
- **Resolution.** IUPAC names go to SMILES with OPSIN³⁶ and RDKit³⁷ canonicalisation (InChIKey, SELFIES³⁸), with a PubChem³⁹ fallback for trivial names; handling open-access label conventions and narrative artefacts raised structure coverage from 24% to 35%.

2.3 Contents and licensing

Table 1 summarises contents and provenance.

The pools differ: PMC records are author-transcribed (median 9 bands), Chemotion records peak-picked from deposited spectra (median 39); users training on band density should treat them apart. `scripts/split_license_pools.py` separates them on `source_doi` and stamps each record’s licence.

Extraction fidelity is measured. On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`) we re-fetched each article and checked every recorded wavenumber against its text: 560/560 bands and 60/60 records were confirmed (Wilson 95% CI 99.3–100% and 94–100%). This bounds *transcription* error — hallucinated, mis-parsed or unit-mangled values — below 1% of bands. It does not measure whether the parser found every IR string in every paper — a recall question requiring human reading; that audit is prepared but not yet run (§8).

`irexp_resolved` (43,060 records, 100% structure-linked) is the benchmark-ready split, ~6× the 6,833-molecule set used to train Spectro¹ (Fig. S2).

3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw *IRSpectra-Bench*, 194 blind elucidation problems, each giving the *molecular formula* (as from HRMS), the *IR band list* and the ¹H and ¹³C *shift lists*; no name, SMILES or scaffold hint. In 10 of the 194 the authors’ peak assignments name a ring system (2CH₂-pyrrolidine, pyridazinone H5). Those compounds are solved 1/10 against 54/184 (29.3%) for the rest (`scripts/prompt_leakage.py`), so the annotation does

Table 1. IRExp dataset contents and provenance.

field	value
experimental IR band-list records	121,233
...co-reporting ^1H and/or ^{13}C NMR	87,075 (72%)
...with a resolved 2D structure	43,060 (35.5%)
...of these, with ^1H and/or ^{13}C NMR	40,702
...of these, with both ^1H and ^{13}C (full quadruples)	33,201
by provenance: author-transcribed from PMC-OA text (CC-BY)	119,345
by provenance: peak-picked from Chemotion ELN deposits (CC-BY-SA)	1,888

not carry the headline; we keep them rather than drop them. Main-round ground truths are spectrally validated by an automated RDKit check (^{13}C peaks vs symmetry-unique carbons, formula match, SELFIES round-trip) excluding merged or incomplete spectra — 6/140, leaving 134; `scripts/validate_benchmark.py` applies the same filter to all three rounds, regenerating every `clean_qids.json` from the released questions and answers.

The filter does not gate on ^1H . In 13 of the 194 retained records the reported ^1H integral exceeds the reference structure’s hydrogen count — residual solvent, water, exchangeable protons, or a rotamer mixture — printed as a diagnostic, not excluded; the cohort was fixed in advance, and re-filtering post hoc on a criterion chosen after seeing results is a degree of freedom a benchmark should not take. We report the sensitivity instead: dropping all 13 moves the headline from 28.4% to 29.3% (53/181), +0.9 points, leaving every conclusion unchanged. The controlled rounds’ 60 compounds are fixed, pre-registered sets, audited the same way but reported rather than filtered (57/60 pass; §8).

Difficulty is declared in advance, as a property of the compound. By RDKit ring analysis a compound is *simple* iff it has at most two rings, no fused/spiro/bridgehead system and ≤ 22 heavy atoms; all others are *complex* (98 simple / 96 complex). The threshold is not load-bearing: sweeping it from 18 to 26 leaves the simple-minus-complex top-1 gap inside a 36–40-point band throughout (36.1 at the narrowest, 39.6 as released; `scripts/difficulty_sensitivity.py`). This axis is distinct from the continuous size gradient of §4.1 (≤ 15 / 16–25 / > 25 heavy atoms), and InChIKey de-duplication across rounds prevents leakage.

The 50/50 balance is by design. The sampler fills each difficulty stratum to half the round; the eligible corpus is 17.5% simple / 82.5% complex. Reweighted to that composition, top-1 falls 28.4% $\rightarrow$ 15.2% [10.6–20.4] and recall 33.5% $\rightarrow$ 19.8% [14.4–25.8]; verification precision must be reweighted too (89.2% $\rightarrow$ 71.5% [50.2–92.5]). We report the benchmark figure as released and the reweighted figure as the better estimate on an arbitrary paper.

Scoring is mechanical. A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference: we score *constitution*, so correct constitution with wrong stereochemistry counts as correct. The *full*, stereochemistry-sensitive InChIKey gives 21.1% top-1 and 25.8% recovered (41/194 and 50/194) against 28.4% / 33.5% — 7.3 points lower on top-1 and 7.7 on recovered — so the constitution figure is an upper bound on full-stereochemistry accuracy. Constitution is the headline because 1D $^1\text{H}/^{13}\text{C}$ /IR rarely fixes absolute configuration: only 10.3% (20/194) of answers carry a *defined* (assigned R/S) stereocentre, and a

model is penalised there for information the prompt never contained (`scripts/score_main.py --stereo`).

Metrics. *Top-1 (exact constitution)*: best-ranked candidate matches at the connectivity layer; *recovered (top-3)*: reference among up to three returned candidates, the lenient “recovery” protocol of ref.². *Generation recall*: reference in the candidate pool *before* re-ranking, the ceiling any verifier can reach (coincides with recovered except in §5.3, where the pool is larger). *Verification precision (conditional on recall)* over recall-positive compounds isolates verification from generation. The forward-verifier ranks by symmetric *chamfer distance* between predicted and observed ^{13}C peak sets (lower is better); Morgan(2, 2048) Tanimoto⁴⁰ gives a graded scaffold-family signal. CIs are bootstrap 95%; model-vs-model differences use McNemar’s exact test⁴¹ with Holm correction⁴².

Solvers are frontier LLM agents (Claude Opus), one sub-agent per batch, closed-book under a consumer subscription: no tools beyond an RDKit formula check, no ground truth, verified by grep-auditing transcripts.

Formula adherence is high but imperfect. Of the 126 candidates carried through forward-verification on the original arm (§5.2), 91.3% (115/126) match the given formula exactly, and 76.6% across the full top-3 pool. On top-1 answers adherence is 95.0% (38/40) in v3 and 90.0% (18/20) in the v2-control — the two pre-registered controlled rounds, 60 compounds between them — against 77.6% (104/134) in the headline main round. Wrong-composition answers score as misses (~one main-round compound in five), so this inflates nothing; the main-round arm is the weakest-constrained (`scripts/analyze_misses.py`).

4. How well do LLMs elucidate real structures?

4.1 Headline performance

Solver agents work blind from formula + IR + ^1H + ^{13}C , returning up to three ranked candidates in bounded contexts reset every 2–12 compounds rather than one long context — the variable §4.3 shows what matters. The benchmark comprises all 194 compounds (134 spectrally-validated + 60 controlled-round; `data/benchmark_main/raw/`).

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	28.4% [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto ≥ 0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

Table 2. Headline elucidation performance on IRSpectra-Bench (n=194). Overall and by difficulty stratum, with bootstrap 95% CIs.

A strictly-validated subset (134 main-clean + 57 controlled-clean = 191) reproduces this within ≈ 1 point on every metric — top-1 28.8%, 95% CI 23–36; recall 34.0%; simple 49.0% / complex 8.4% — so the asymmetric inclusion of pre-registered controlled sets does not drive it. The deliberate 50/50 difficulty balance does: reweighted to the 17.5%-simple composition of the eligible corpus, top-1 is 15.2% [10.6–20.4] and recall 19.8% [14.4–25.8] (§3). Per-stratum figures are unchanged; the reweighted number is what to read as performance on an arbitrary paper. Accuracy falls monotonically with size (Fig. 2): 60.5% top-1 at ≤ 15 heavy atoms, 28.3% at 16–25, 7.0% above 25 (Fig. S3).

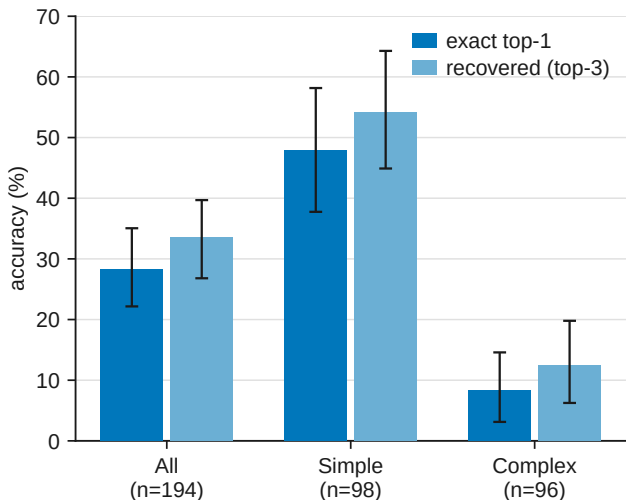


Fig. 2. Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194), with bootstrap 95% confidence intervals. The benchmark separates a realistic difficulty range: simple targets are solved four to six times as often as complex ones on both metrics.

Of the 137 analysable top-1 misses (139 in all; two predictions did not parse; `scripts/analyze_misses.py`), 76.6% are constitutional isomers of the true structure against 23.4% with the wrong formula — not mostly regiochemistry: only 22.6% share the true Murcko scaffold, 2.9% reach Tanimoto ≥ 0.85 , and the median isomeric-miss Tanimoto is 0.39. Strict regiochemistry accounts for a fifth of failures; most misses are larger rearrangements of the same atoms. Near-degenerate $^1\text{H}/^{13}\text{C}$ shifts under-determine connectivity; the misses show *wrong connectivity at the right composition*.

4.2 Reconciling with prior reports

Our 28% top-1 sits far below $\approx 100\%$ on curated NMR-only demos². Four choices inflate prior reports — difficulty, lenient top-3 scoring, starting-material hints, hand curation — and we name them without apportioning the gap. Trained baselines on simulated or NIST gas-phase IR reach 48–94% in-distribution^{1,6,18}; ours is 28.4% on blind literature spectra. On

≤ 15 -heavy-atom compounds our 60.5% approaches Alberts’ 63.8% (IR-only, single instrument), but settings differ on every axis — modality, spectrum realism, library homogeneity — so neither number bounds the other. High in-distribution scores overstate real-world performance; the $\approx 40\times$ MolPuzzle swing for one model (§1.1) bounds what unaudited near-100% claims can bear.

4.3 Methodology dominates: a within-compound control

The same 20 molecules were solved two ways: (a) one long context, no tools; (b) four agents, five compounds each, with RDKit formula-checking. Recovered rose 5% $\rightarrow$ 15% and top-1 0% $\rightarrow$ 15% (McNemar $p=0.25$ at $n=20$). Bounded, frequently-reset contexts with tool access may raise measured performance — a directional demonstration, not a quantified contribution to the headline gap.

4.4 Model comparison: the benchmark orders capability but separates only the extremes

We solved a fixed 24-compound subset blind with four Claude models spanning a wide capability range, including the newest (Table 5), under one prompt, one scorer and one candidate budget (Fig. 3, Table 3).

One protocol asymmetry must be disclosed: the three comparison models solved the 24 compounds as four six-compound contexts each, whereas the Opus column reuses the headline run, where those items sat in one six-compound and two twelve-compound contexts — the variable that §4.3 measures at 5% $\rightarrow$ 15%. The Opus estimate is therefore not protocol-matched and, if anything, handicapped by the longer contexts; this touches neither the nesting nor the Fable-vs-Haiku contrast, and a clean re-run is outstanding in docs/MODELS.md.

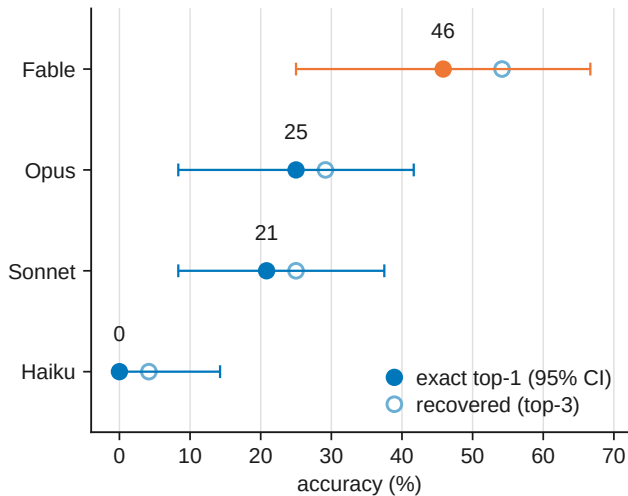


Fig. 3. Four-model comparison on a fixed 24-compound subset, solved blind under one protocol. Outcomes are strictly nested — each stronger model solves a superset of the weaker one’s compounds — so the benchmark is capability-sensitive, but at $n=24$ it is underpowered to separate adjacent models (§4.4).

Table 3. Four-model comparison on a fixed 24-compound subset. All four models ran the identical blind protocol.

CIIs are bootstrap except Haiku’s: Clopper–Pearson exact for 0/24, where the percentile bootstrap is degenerate.

Outcomes are strictly nested (Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable); only Fable-vs-Haiku reaches significance (Holm $p=0.006$). Opus headline runs used longer contexts than the comparison models (§4.3).

4.5 Domain case study: battery-electrolyte chemistry

IRSpectra-Bench-Electrolyte (46 scored compounds across six electrolyte functional classes, held out and solved blind) matches the headline regime: top-1 26%, recovered 28%; the true structure enters the pool for 13/46 and ranks first in 12/13 — recall-bound, not domain-specific (Fig. S4, Table 4). Per-class intervals overlap (χ^2 $p\approx 0.56$); $\text{sp}^3\text{-C-F}$ reaches 50%, sulfonyl and nitrile 12%.

Intervals overlap throughout (six-class χ^2 $p\approx 0.56$; best-vs-worst Fisher exact $p\approx 0.28$), so the ordering is hypothesis-generating, not established — C–F couplings pin $\text{sp}^3\text{-C-F}$ regiochemistry, while sulfonyl and nitrile targets turn on oxidation-state ambiguity and heteroaromatic substitution that $^1\text{H}/^{13}\text{C}$ shifts underdetermine. It is the recall-bound failure of §4.1 in a domain-specific guise, and it is where §5’s *forward-verification* recipe plays a role *analogous* to (not a replacement for) computational-NMR validation.

4.6 Is the model reading the spectra? A formula-only control

We reran the blind protocol with spectra masked, leaving only the formula (⁴³).

Table 5. Formula-only control. Paired on the same 60 compounds as §5.

Outcomes are perfectly nested (McNemar $p=0.001$). Accuracy is flat in source-paper year across all 194 ($r=-0.007$; Fig. 4), bounding pretraining recall. We claim a strong bound, not exclusion (§8).

4.7 Does the diagnosis hold outside one vendor? A four-vendor replication

Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol solved the same 60-compound arm under the identical protocol (Table 6). Verification precision exceeds generation recall in every arm (Fig. 5); bootstrapping separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional. Three models beat Claude on recall; candidate budgets differ (ours 2.20 vs 3.00 per compound), so recall rankings are approximate. A clean-clone control for Grok (recall 28/60 vs 32/60, McNemar $p=0.39$) bounds key leakage. Models ran through a coding-assistant harness with undisclosed reasoning tiers (§8).

Table 6. Cross-vendor decomposition on the 60-compound arm. Recall and precision have different denominators, so the criterion is the inequality rather than a difference.

5. Forward-verification elucidation

5.1 Method

The inverse direction is the model’s hard, isomer-blind direction; the forward direction (structure $\rightarrow$ spectrum) is its easy, accurate one². We close a training-free generator–verifier loop:

generate candidate structures (inverse) $\rightarrow$
forward-predict each candidate’s ^{13}C spectrum,
blind to the observed spectrum $\rightarrow$ *re-rank* candidates by the distance between predicted and observed $^{13}\text{C} \rightarrow$ return the best match.

The inverse solver’s 373 deduplicated candidates across 194 targets are forward-predicted in shuffled, anonymised batches, blind to the observed spectrum and target identity. Predicted and observed ^{13}C peak sets are compared by symmetric chamfer distance (Fig. 6).

Regioisomers separate by a median 1.21 ppm in forward prediction, but 82% lie within the predictor’s ≈ 2 ppm error — a thin margin confirmed against a derangement null at $p=0.001$ (§5.5). Sharper predictors (GNN, DFT-coupled models^[44,45,46]) would move the verification ceiling, not search.

5.2 Result

The first column below is the 60-compound arm (v3 + v2-control) on which §5.3–§5.5 build; the second, the full benchmark. All 373 candidates were forward-predicted — nothing here is a lower bound; the self-ranking row re-derives Table 2 from §4’s output.

Table 7. Forward-verification decomposition. Original arm and full benchmark.

When the true structure is among the candidates, forward-verification selects it in 58 of 65 cases (89%). Of the 65, 28 had a single candidate — scored by construction by any ranker — so the pooled figure is not the verifier’s margin over chance. Where a choice existed ($n=37$), forward-verification gets 30/37 (81%) against a 54.0% derangement floor (§5.5) — +27.1 points; self-ranking 27/37 (73%).

The margin over self-ranking remains small and unresolved: seven compounds gained, four lost, McNemar exact $p=0.55$. We do not claim it. Precision is high in absolute terms while recall binds: top-1 moves only 28% $\rightarrow$ 30% because the true structure was never proposed for 129 of 194 compounds, which no re-ranking can repair. The verifier converts recall almost completely on *simple* targets (50/53, 94%) and ties self-ranking on *complex* ones (8/12, 67%). Elucidation factorises into two near-independent levers: a strong verifier and a generator at 34% recall that is the wall (Fig. 1).

5.3 Generate-wide: testing the recipe

The decomposition implies *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling⁴⁷. Ten solver agents proposed up to six regiochemistry-aware candidates per compound, pooled with the originals and re-ranked.

model	top-1	recovered	top-1 95% CI
Claude Fable 5	46%	54%	[25–67]
Claude Opus	25%	29%	[8–42]
Claude Sonnet	21%	25%	[8–38]
Claude Haiku	0%	4%	[0–14]

Table 4. Per-class performance on IRSpectra-Bench-Electrolyte (n=46).

electrolyte class	n	top-1	95% CI	recovered (top-3)	95% CI
sp ³ -C–F	8	50%	[22–78]	50%	[22–78]
carbonate	7	29%	[8–64]	43%	[16–75]
phosphoryl	8	25%	[7–59]	25%	[7–59]
glyme / oligoether	7	29%	[8–64]	29%	[8–64]
sulfonyl / sulfonate	8	12%	[2–47]	12%	[2–47]
nitrile	8	12%	[2–47]	12%	[2–47]

condition	top-1	recovered (top-3)
formula only	3/60 (5%)	3/60 (5%)
formula + IR + ¹ H + ¹³ C	14/60 (23%)	19/60 (32%)

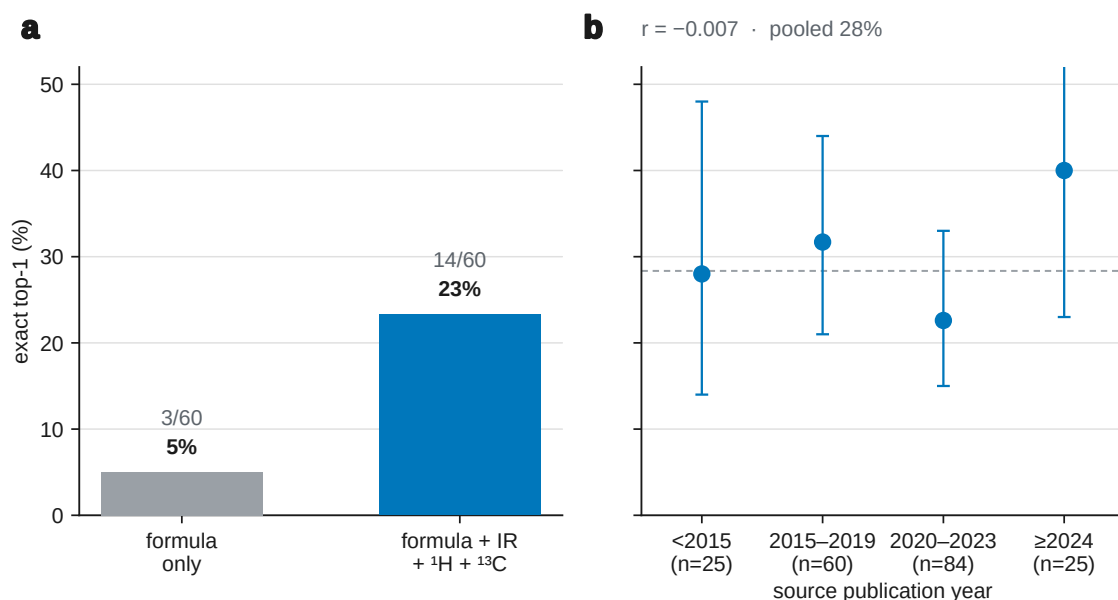


Fig. 4. Two contamination controls. (a) Formula-only vs full modality on 60 compounds. (b) Accuracy vs source-paper year (n=194).

model	generation recall	verification precision recall	multi-candidate only
Claude Opus (Table 7)	19/60 = 32% [21–44]	16/19 = 84% [62–94]	10/13 = 77%
Grok 4.6	32/60 = 53% [41–65]	20/32 = 62% [45–77]	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50% [38–62]	22/30 = 73% [56–86]	22/30 = 73%
GPT-5.6 Sol	25/60 = 42% [30–54]	17/25 = 68% [48–83]	16/24 = 67%

	60-compound arm	full benchmark (n=194)
generation recall	19/60 (32%)	65/194 (34%)
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58/194 (30%)
<i>conditional on recall</i> — self-ranking	14/19 (74%)	55/65 (85%)
<i>conditional on recall</i> — forward-verification	16/19 (84%)	58/65 (89%)
...single-candidate compounds within that set	6/19	28/65
<i>multi-candidate only</i> — self-ranking	8/13 (62%)	27/37 (73%)
<i>multi-candidate only</i> — forward-verification	10/13 (77%)	30/37 (81%)

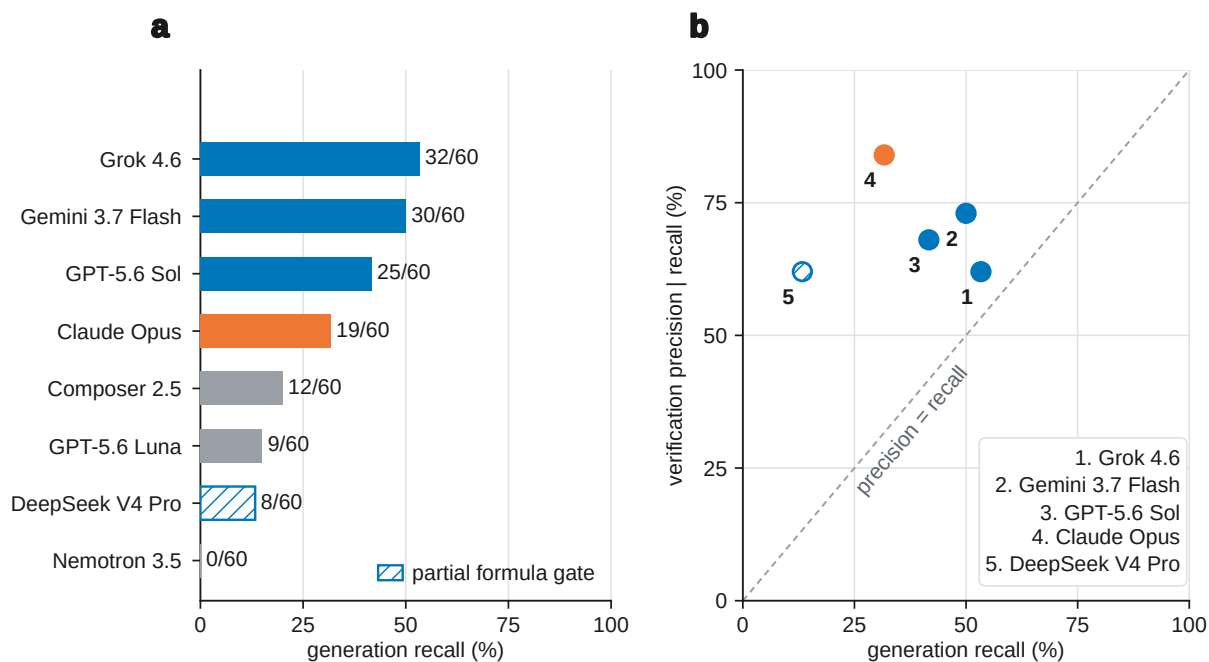


Fig. 5. Cross-vendor decomposition on the 60-compound arm: generation recall vs verification precision (§4.7).

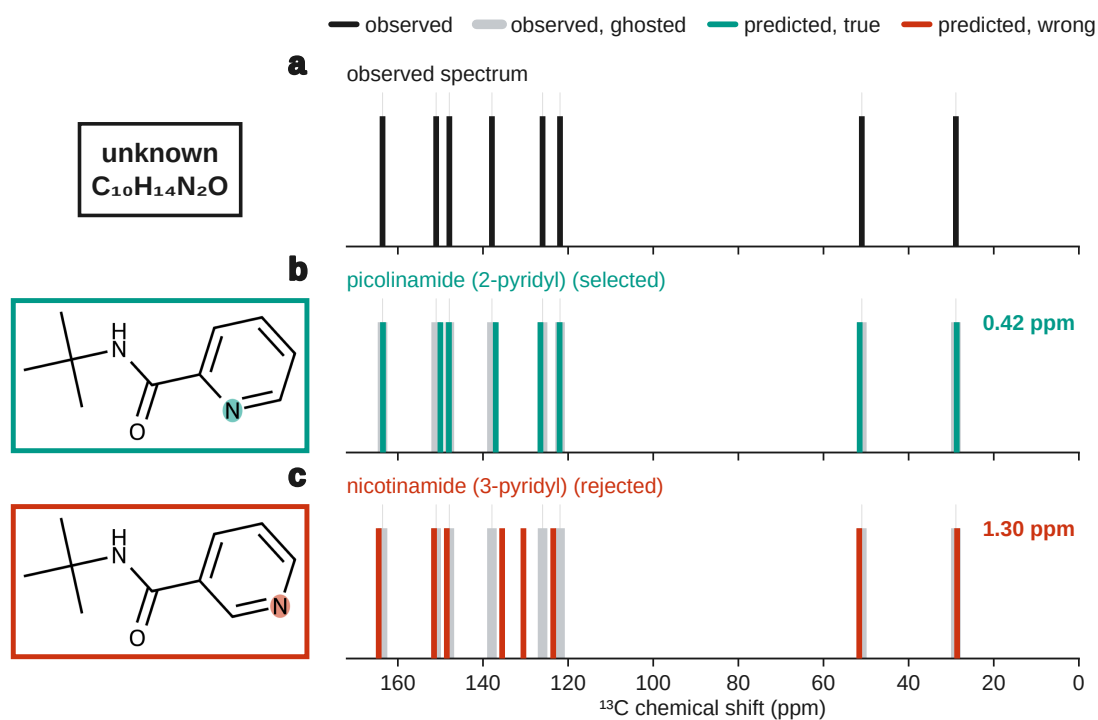


Fig. 6. Forward-verification on a benchmark regioisomer pair: picolinamide and nicotinamide are indistinguishable to the inverse task, but forward-predicted ^{13}C sticks separate the true isomer from the alternative (§5.1).

Table 8. Generate-wide vs original. On the 60-compound arm only, since wide generation was run there; the “original” column is Table 7’s first.

Wide generation lifts recall 32% $\rightarrow$ 42% and top-1 23% $\rightarrow$ 30% (McNemar $p=0.34$; Fig. 7). Recall plateaus at 42% on polycyclic targets; verification precision falls 84% $\rightarrow$ 72% — the training-free ceiling. Roughly a third of misses need only regiochemistry around a correct scaffold; two thirds need a scaffold never proposed.

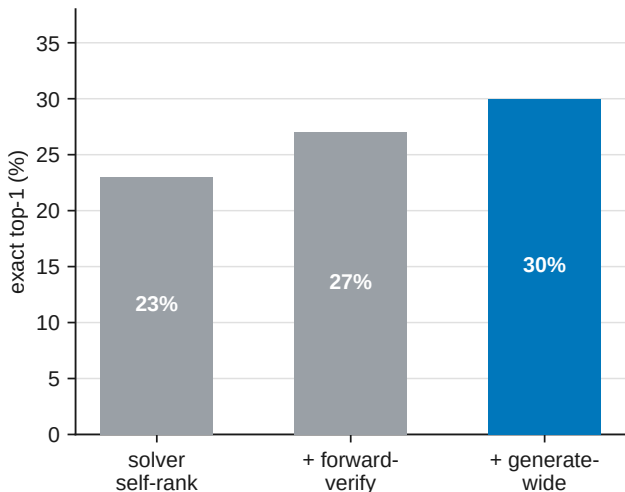


Fig. 7. Forward-verification inference ladder on the 60-compound arm (§5.3). Each stage adds a check on the same compounds; the hero bar is generate-wide top-1.

5.4 Non-LLM verifiers: a deterministic lookup and a learned model

§5.3 suggests replacing the verifier with a non-LLM ^{13}C predictor. We tested a HOSE-code⁴⁸-style lookup and a small message-passing GNN, both trained on the same nmrshiftdb2 dump⁴⁹ and applied to the same §5.2 candidate sets so only the predictor changes. The GNN is deliberately modest; sharper purpose-built ^{13}C models exist^{44,45,50}.

The HOSE lookup ties self-ranking; the GNN reaches 59/65 (91%) vs the LLM’s 58/65 (89%), directionally (Table 9, Fig. S6). Wagen’s MagNET-in-the-loop study^{51–53} moves connectivity 46.4% $\rightarrow$ 55.4% on eight problems — same order as our ladder, underpowered at $n=8$.

5.5 Negative control

Deranging observed ^{13}C pairings collapses verification precision 89.2% $\rightarrow$ 73.8% mean (one-sided $p=0.001$); on multi-candidate sets alone, 81.1% vs a 54.0% floor (+27.1 points). Chamfer margin does not support selective prediction (flat top-1 across coverage fractions) — a reported null.

5.6 Is the recall wall task-intrinsic? A trained-generator probe

That ceiling is a property of training-free LLM elicitation, not the task. Pooling a small IReXp-fine-tuned $^1\text{H}/^{13}\text{C} \rightarrow \text{SMILES}$ generator with Claude’s candidates lifts recall 33.5% $\rightarrow$ 54.1% and top-1 28.4% $\rightarrow$ 35.1% (McNemar $p=0.015$; Fig. S5). Zero-shot without IReXp fine-tuning recovers 0/248; with it, 25%. NMR-Solver (52.9%)²⁷ and trained IR transformers (63.8%)¹⁸ confirm the task is not bound at 28–30%.

6. The decomposition across the published literature

The split measured above applies to any system reporting top-1 = a and top- k = b : recall $\geq b$, conditional precision $\leq a/b$. Table 10 applies this to every system we could read in full (scripts/literature_decomposition.py).

Table 10. Recall and ranking loss from published top- k figures. Rows are comparable only within a correctness criterion; stereochemistry-strict and connectivity figures differ by 21 points on NMR-Solver’s identical predictions. Candidate budgets are given because recall over three candidates is mechanically smaller than over ten.

Three groups changed only the data and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse. Published gains come mainly from candidate supply (NMRAgent ablation²⁸, NMRGym formula ablation¹⁹). Recall binds where spectra are real and heterogeneous (87–91% of our loss); ranking dominates on simulated or single-library data (38–39%). The field almost never reports whether the true structure was proposed at all (¹⁶ excepted).

7. Discussion

Frontier LLMs are good verifiers (89% conditional precision) and weak proposers (34% recall) on real literature spectra. The contribution is a diagnosis with a bounded, training-free improvement, not a solved elucidator.

The split held across four Claude models, a battery-electrolyte subset, four verifiers and four vendor families (§6). Concurrent work sharpens it: Kliuev *et al.* place the best LLM at 24% on 105 peak lists²⁹; Espejo Morales *et al.* show environment design moves proposal (71.1% on 15 molecules) while their education $\rightarrow$ industrial drop (80.9% $\rightarrow$ 20.6%) is our claim measured independently³⁰; Wagen’s MagNET loop^{51,52} moves connectivity 46.4% $\rightarrow$ 55.4% on eight problems. Those three levers — proposal environment, external simulation, wider generation — attack one search problem.

Training is not foreclosed: IReXp fine-tuning lifts recall to 54% (§5.6). What compounds is open experimental data, honest benchmarks, and inference scaffolding that rides each new model. Two practical findings: bounded contexts with tool access may help (§4.3); use forward-predicted ^{13}C agreement to re-rank, not as an abstention gauge (§5.5). Learned embeddings rarely beat ECFP under controlled evaluation⁵⁴.

8. Limitations

Scoring is mechanical; solver runs were transcript-audited for zero ground-truth access (transcripts on request). **(i) Pre-training contamination** is bounded by formula-only (23% $\rightarrow$ 5%) and flat recency controls (§4.6) but not excluded; verbatim spectral strings in prompts remain a retrieval confound. **(ii) Human audit** of elucidation outputs is prepared (data/audit/) but not yet run. **(iii) Headline $n=194$ is Claude Opus**; cross-vendor evidence is on 60 compounds (§4.7). **(iv) Battery subset** uses literature electrolyte chemistry, not operando spectra. **(v) Constitution-only scoring** (21.1% with stereochemistry). **(vi) Single-sample scoring** per compound. **(vii) Organic literature bias.** **(viii) Author-transcribed peak lists**, not raw instrument files — a different regime from Espejo *et al.*³⁰.

	original (60-compound arm)	generate-wide
generation recall (true structure among candidates)	32%	42%
forward-verified top-1	27%	30%
verification precision (conditional on recall)	84%	72%

Table 9. Verifier comparison, conditional on recall.

verifier	60-compound arm (n=19)	full benchmark (n=65)	held-out ¹³ C MAE
solver self-ranking	14/19 (74%)	55/65 (85%)	—
deterministic HOSE <i>lookup</i>	14/19 (74%)	55/65 (85%)	3.23 ppm
learned GNN (same data)	16/19 (84%)	59/65 (91%)	1.70 ppm
LLM forward-verifier (§5.2)	16/19 (84%)	58/65 (89%)	—

system	data (n)	top-1	top-k (k)	recall	prec.
scored on connectivity					
this work, solver alone	literature (194)	28.4%	33.5% (3)	33.5%	84.6%
this work, + forward-verification	literature (194)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver ²⁷	literature (450)	52.9%	67.3% (10)	≥67.3%	≤78.6%
NMRAgent ²⁸	literature (450)	61.6%	70.0% (10)	≥70.0%	≤88.0%
scored with stereochemistry					
this work	literature (194)	21.1%	25.8% (3)	≥25.8%	≤81.8%
NMR-Solver ²⁷	simulated (1,000)	66.9%	89.9% (10)	≥89.9%	≤74.4%
NMR-Solver ²⁷	literature (450)	31.6%	53.8% (10)	≥53.8%	≤58.7%
Espejo Morales ³⁰	education (236)	80.9%	90.0% (5)	≥90.0%	≤89.9%
Espejo Morales ³⁰	industrial (34)	20.6%	29.1% (5)	≥29.1%	≤70.9%
exact match, stereo handling not stated					
Alberts, 6–13 atoms ¹⁸	NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
Alberts, 5–35 atoms ¹⁸	NIST IR (5,024)	59.9%	78.5% (10)	78.5%	76.4%
SpecX, random split ²⁰	simulated (99,439)	59.0%	81.8% (10)	81.8%	72.2%
SpecX, scaffold split ²⁰	simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%
IR-Agent ¹⁵	NIST IR (905)	10.3%	21.6% (10)	21.6%	47.7%
criterion not stated					
Priessner ¹⁶	experimental (34)	20.6%	—	26.5%	77.8%

9. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`, parsed deterministically, and resolved with OPSIN, RDKit and SELFIES, with an optional cached PubChem fallback.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, and deduplicated across rounds by InChIKey. A compound was admitted only if its raw ^1H payload — multiplicities and J values as printed — could be re-extracted verbatim from the PMC-OA full text of its source article (`benchmark_v2.raw_1h_for`), which puts genuine coupling information in the prompt and, unavoidably, the source article’s exact string (§8). The battery-electrolyte subset (§4.5) was drawn by SMARTS filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile, $\text{sp}^3\text{-C-F}$, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding compounds used elsewhere; it was J -enriched and spectrally validated. Solver and forward-prediction agents were independent Claude-Opus sub-agents under the same subscription, instructed closed-book and audited for zero web/answer access. Scoring used RDKit InChIKey-connectivity match; similarity Morgan(2, 2048) Tanimoto; forward-verification a symmetric chamfer distance over ^{13}C peak sets. The core protocol trains no model and uses no paid API; the §5.6 generator and the §5.4 learned ^{13}C verifier are the only trained components, reported as complements and fenced from the headline results.

Models and versions. Headline runs used Claude Opus via consumer subscription (2026-06-09–11); formula-only control 2026-07-28; forward-prediction extension 2026-08-07. No model snapshot or decoding parameters can be reported — the harness exposes neither (§9). Fixed are frozen per-compound outputs and mechanical scorers (`docs/MODELS.md`).

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions and scorer outputs are released; the sampler, scorer and forward-verification harness are scripted end-to-end.

Supporting Information figures

These are supplied as a separate Electronic Supplementary Information document (`docs/paper_esi.pdf`, built by `scripts/build_pdf.py`); they are listed here for reference.

- **Fig. S1** (`docs/figures/fig0_overview.png`) — study design: open multimodal data (IRexp) $\rightarrow$ blind, complexity-stratified benchmark $\rightarrow$ decoupled blind solving $\rightarrow$ forward-verification re-ranking; training-free core pipeline.
- **Fig. S2** (`docs/figures/fig4_dataset.png`) — IRexp composition: IR records $\rightarrow$ NMR-paired $\rightarrow$ structure-linked $\rightarrow$ full IR + ^1H + ^{13}C + structure quadruples.
- **Fig. S3** (`docs/figures/fig2_size.png`) — accuracy vs molecular size (heavy-atom bucket); the monotonic 60% $\rightarrow$ 7% top-1 gradient.
- **Fig. S4** (`docs/figures/fig6_electrolyte.png`) — top-1 and recovered accuracy on IRSpectra-Bench-Electrolyte by battery-electrolyte chemical class ($n=46$): $\text{sp}^3\text{-C-F}$ easiest (50%), sulfonyl and nitrile hardest (12%); overall 26%/28%.
- **Fig. S5** (`docs/figures/fig_generator_probe.png`) — trained-generator probe (§5.6; complement, not training-free protocol); generation recall and deterministic-HOSE

top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator. Enumeration’s near-degenerate isomers collapse the verifier (28.4 $\rightarrow$ 16.0%) while the generator’s formula-correct candidates convert (28.4 $\rightarrow$ 35.1%).

- **Fig. S6** (`docs/figures/fig_verifier.png`) — learned-verifier probe (§5.4; complement, not training-free protocol). (A) Conditional-on-recall top-1 ($n=65$, whole benchmark) for four verifiers: a GNN trained on the same `nmrshiftdb2` data as the HOSE lookup reaches the LLM verifier’s level (91% vs 89%) where the lookup (85%) does not move off the solver’s own ranking. (B) Held-out ^{13}C MAE — the learned model is $\approx 2\times$ sharper (1.70 vs 3.23 ppm).

Author contributions

I.Y.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft. **R.S.:** methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, §5.4 and §5.6), writing — review and editing. **R.A.V.-H.:** conceptualization, methodology, supervision, writing — review and editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

Data, predictions, scoring scripts and figure regeneration are in the project repository; Table 11 lists each component. A frozen snapshot will be deposited on Zenodo (DOI at proof); GitHub is the development mirror. IRexp is also mirrored at huggingface.co/datasets/ilkhamfy/IRexp (use `data/train_no_bench.jsonl.gz` for fine-tuning without benchmark leakage).

IRexp redistributes extracted numeric data only (band lists, shifts, structures, source DOIs) from PMC-OA (CC-BY-4.0) and Chemotion (CC-BY-SA-4.0); pools are separable by `source_doi` (`scripts/split_license_pools.py`). Code is MIT. Cite the Zenodo deposit and attribute sources via each record’s `source_doi`. De-leak with `contrib/generator_probe/build_exp_manifest.py` before training against the benchmark; withhold `data/audit/key.jsonl` and `data/modality/key.json` from blinded reviewers.

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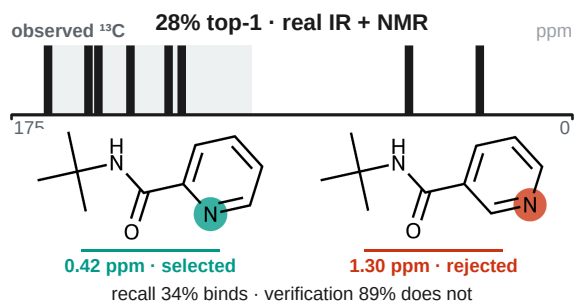
Table 11. Released artefacts, and the data and code behind each one.

component	data and code
IRexp and its structure-complete split	data/irexp/, data/irexp_resolved/ — spectro_scraper/ (mining pipeline)
benchmark rounds and the within-compound control	data/benchmark*/ — scripts/benchmark_v2.py
ground-truth integrity audit	data/benchmark*/clean_qids.json — scripts/validate_benchmark.py
headline accuracy (Table 2)	scripts/score_main.py
battery-electrolyte subset (§4.5)	data/benchmark_electrolyte/ — scripts/build_electrolyte_bench.py, scripts/score_electrolyte.py
formula-only contamination control (§4.6)	data/modality/ — scripts/modality_ablation.py
publication-recency control (§4.6)	data/audit/recency_control.json — scripts/contamination_recency.py
forward-verification, original arm (§5.2)	data/fverify/ — scripts/forward_verify.py
...extended to all 194 compounds	data/fverify_main/ — scripts/forward_verify_main.py, scripts/forward_verify_all.py
generate-wide arm (§5.3)	data/gw/, data/fverify2/ — scripts/score_generate_wide.py, scripts/ladder_significance.py
...its coverage gap, closed	data/fverify_gw/ — scripts/forward_verify_gw.py
non-LLM verifier comparison (Table 9)	data/fverify/hose_results.txt, data/fverify/verifier_table_results.txt — scripts/hose_predict.py (incl. coverage), scripts/verifier_table.py, scripts/verifier_leakage.py
negative control and selective prediction (§5.5)	scripts/verifier_diagnostics.py
what a miss is, and isomer separability (§4.1, §5.1)	scripts/analyze_misses.py, scripts/isomer_separability.py
difficulty-threshold sensitivity (§3)	scripts/difficulty_sensitivity.py
licence-pool split (§2.1)	scripts/split_license_pools.py
recall headroom and scaffold enumeration	scripts/analyze_recall_headroom.py, scripts/enumerate_isomers.py, scripts/closing_the_gap.py
blinded expert-audit package (§8)	data/audit/ — scripts/make_audit_sample.py
corpus reweighting (§3)	scripts/corpus_reweight.py
cross-vendor recall at matched budget, and the paired gap (§4.7)	data/cross_vendor/ — scripts/cross_vendor_budget.py, scripts/cross_vendor_gap.py
ring-system names in peak assignments (§3)	scripts/prompt_leakage.py
paraphrase-invariant benchmark, for the control not yet run (§8)	— (regenerated, not deposited) — scripts/jitter_control.py
manuscript integrity gates	scripts/check_manuscript.py, scripts/verify_statistics.py, scripts/check_compression.py, scripts/check_layout.py
all figures (PNG + PDF twins)	docs/figures/ — scripts/make_all_figures.sh
trained generator probe (§5.6)	contrib/generator_probe/, data/fverify_gen/
learned GNN verifier (§5.4)	scripts/gnn_predict.py, data/nmrshiftdb/gnn_c13.pt, data/fverify/gnn_results.txt

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IRSpectra-Bench and IRExp on real literature IR + ^1H + ^{13}C :
candidate recall, not verification, limits LLM elucidation —
the true structure is proposed for 34% of 194 compounds and,
once proposed, selected 89% of the time, lifting top-1 from
28% to 30%.